CSIT375 AI for Cybersecurity

Assignment

Due Date: 11pm, Dubai Time, 13 Nov. 2026

This task focuses on the implementation of AI/ML algorithms in the context of Cyber Security. Please refer to the subject outline (Assignment 2) for complete details and requirements. The weightage of this assessment is 20%.

You need to upload your group details to moodle (check the Group formation section) by 12th Oct. 2025 so that I can assign you a group ID.

A few suggested implementation ideas are provided below; however, you are welcome to propose your own idea, provided it aligns with the scope of CSIT375.

Kindly **prepare and send your project proposal** for approval before proceeding with the implementation.

Instructions:

- **You can use Python to implement the AI/ML algorithms**
- Your submission should be a **zip file that includes implementation (code) file along with a word document (report)**. The word doc should include readme content along with output samples and results.
- In the word doc, include following sections- Introduction of the project, Discussion of the implemented ML/AI model, Results, Discussion and references. The report should be precise and there is no word limit.

Marking Rubrics:

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Marks
Understanding of Concepts	Demonstrates thorough understanding of all relevant concepts; ideas are clear and accurate.	Shows good understanding; minor gaps in explanation or accuracy.	Basic understanding; some significant gaps or misunderstandings.	Limited or incorrect understanding of concepts; major errors.	/25
Analysis & Problem Solving	Excellent analysis; applies concepts creatively to solve problems; insights are original and well-justified.	Good analysis; applies concepts correctly; minor gaps in reasoning.	Basic analysis; limited application of concepts; reasoning sometimes unclear.	Poor analysis; concepts not applied correctly; reasoning weak.	/25

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Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Marks
Organization & Structure	Assignment is very well-organized, logically structured, and easy to follow; headings, sections, and flow are clear.	Generally organized; minor issues in flow or structure.	Organization is basic; ideas sometimes unclear; flow inconsistent.	Poorly organized; hard to follow; lacks logical structure.	/15
Research & References	Uses extensive, relevant, and credible sources; references are correctly formatted.	Uses relevant sources; minor referencing errors.	Limited sources; some relevance issues; referencing inconsistencies.	Inadequate or irrelevant sources; referencing missing or incorrect.	/15
Presentation & Clarity	Clear, concise, and professional presentation; free from grammatical/spelling errors.	Generally clear; few minor errors.	Some clarity issues; several errors present.	Poor presentation; many errors; difficult to read.	/10
Collaboration (if group)	Effective collaboration; roles well-defined; all members contribute equally.	Good collaboration; minor imbalance in contributions.	Basic collaboration; unequal contributions noticeable.	Poor collaboration; one or few members did most work.	/10

- Don't forget to include a team member's contribution sheet.
- You can be five students in a group. The groups could be from any lab or lecture.
- Keep report submission name as "**StudentgroupID-CSIT375-report.zip"** and submit it via Moodle by the due date.
- You need to record the demonstration of your project and submit the recorded video on the link provided on Moodle.

Total Marks: 100

Code + Report: 80 marks

Recorded Demonstration: 20 marks (The recorded demo should showcase your implementation and output.)

- Plagiarism should not be tolerated, check the UOW plagiarism policy.
- For any queries you can discuss with me.

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- **There should be no Gen-AI tools generated plagiarism in your programs**. In case student use AI tools then-the students must declare that they have used generative artificial intelligence, specifically (name of the AI in question) in the writing of the report and/or in the creation of images, graphics, tables, codes or their corresponding captions (please specify in all cases where it has been used). The penalty will be decided by the examiner and depends on the percentage of the use of Gen_AI in this assignment.

Project 1: AI-Powered Malware Detection and Classification System

Objective: To develop an AI-powered system that detects and classifies malware in real-time by analyzing incoming data files and network traffic. The goal is to prevent malicious attacks and help security teams respond swiftly to new threats. You can use dataset provided during the lab activities.

The implementation should include following phases:

AI Model Development: The security team aims to create an AI model that can automatically classify files and network activities as malicious or benign. If detected as malicious, the model should classify the type of malware.

1. **Machine Learning Techniques:**
    - **Supervised Learning**: Train models on labeled datasets.
        - **Algorithms**: Random Forest, Gradient Boosting, or Support Vector Machines (SVM) for traditional models.
    - **Deep Learning**: Use Convolutional Neural Networks (CNN) or Recurrent Neural Networks (RNN) to learn from malware behavior patterns and sequences.
        - **Algorithms**: LSTMs for dynamic behavior analysis, CNN for static file analysis.
2. **Malware Classification Model:**
    - **Feature Extraction**: Input features such as API calls, permission usage, file structure, and network traffic etc..
    - **Classifier**: A multi-class classifier trained to categorize malware into types (e.g., ransomware, trojan, etc.) or families.
3. **Model Evaluation**:

- **Metrics**: Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to evaluate performance.
- **Cross-Validation**: Ensure robustness by using cross-validation techniques.

Project 2: AI-Powered Phishing Detection

Objective: The project aims to develop an AI-driven system that detects phishing attempts in emails, websites, and messages. It will not only identify phishing content but also prevent users from falling victim by providing real-time alerts and taking automated preventive measures.

The phases under this project are:

AI Model Development

Scenario: An AI model is trained to distinguish between legitimate and phishing emails, URLs, or messages based on various features such as sender behavior, language cues, and suspicious patterns in the URLs.

1. **Various Feature Estimation (depend on the dataset):**
   - **Email Features**:
     - Sender email domain analysis (spoofed domains, unknown domains).
     - Subject line characteristics (common phishing terms, urgency indicators).
     - Body content analysis (presence of links, HTML content, suspicious attachments).
   - **URL Features**:
     - URL length, redirection chains, usage of IP addresses instead of domain names.
     - Lexical analysis (e.g., common phishing terms, domain similarity to well-known brands).
   - **Textual Analysis**:
     - Language models (NLP-based) to detect phishing patterns such as impersonation or scare tactics.
     - N-grams or TF-IDF (Term Frequency-Inverse Document Frequency) features to capture word patterns indicative of phishing.
2. **Model Selection:**
   - **Machine Learning Models**:
     - **Logistic Regression**: A lightweight model for binary classification of phishing vs. non-phishing.

        - **Random Forest or Gradient Boosting**: For handling complex interactions between features.
    - **Deep Learning Models**:
        - **RNNs (LSTMs)**: For capturing long-term dependencies in email content and detecting sophisticated phishing attempts.
        - **Transformer-based Models (BERT, GPT)**: Pre-trained models fine-tuned to classify phishing emails based on contextual understanding of text.
3. **Phishing Classification Model**:
    - **Binary Classification**: The model labels emails or URLs as "Phishing" or "Legitimate."
    - **Multi-class Classification**: The system can also classify phishing attempts into categories (e.g., spear-phishing, bulk-phishing, URL-based, credential harvesting).
4. **Model Evaluation**:
    - **Metrics**: Accuracy, Precision, Recall, F1-Score
    - **Cross-validation**: K-fold cross-validation to avoid overfitting and ensure the model generalizes well.

Phishing Website Detection using Machine Learning

- Train a classifier (e.g., Logistic Regression, Random Forest, or BERT) to identify phishing URLs based on lexical and content-based features.

Spam and Phishing Email Detection using NLP

- Apply Natural Language Processing (NLP) with models like TF-IDF + ML classifiers or transformer models (BERT, DistilBERT) to detect malicious emails.

Ransomware Detection using Behavioral Analysis

- Use ML models to analyze system-level behaviors or process calls to detect ransomware before encryption starts.

Anomaly Detection in IoT Networks

- Apply unsupervised learning (e.g., Autoencoder, Isolation Forest) to detect unusual activity in IoT environments.

Botnet Detection using Machine Learning

- Classify network traffic to distinguish between normal and botnet-related activities using supervised or unsupervised methods.

Secure Log Analysis with NLP

- Use NLP to parse, classify, and detect anomalies in system and network logs automatically.

You can download datasets from below links:

https://www.kaggle.com/datasets/shashwatwork/web-page-phishing-detection-dataset

https://www.kaggle.com/datasets/shashwatwork/phishing-dataset-for-machine-learning

https://data.mendeley.com/datasets/6tm2d6sz7p/1

https://github.com/GregaVrbancic/Phishing-Dataset

https://www.kaggle.com/datasets/blackarcher/malware-dataset

https://www.kaggle.com/datasets/amauricio/pe-files-malwares

https://www.kaggle.com/datasets/sid321axn/malicious-urls-dataset

Or can use data files provided during the labs.